Dr. AI v1.0 Database ERD

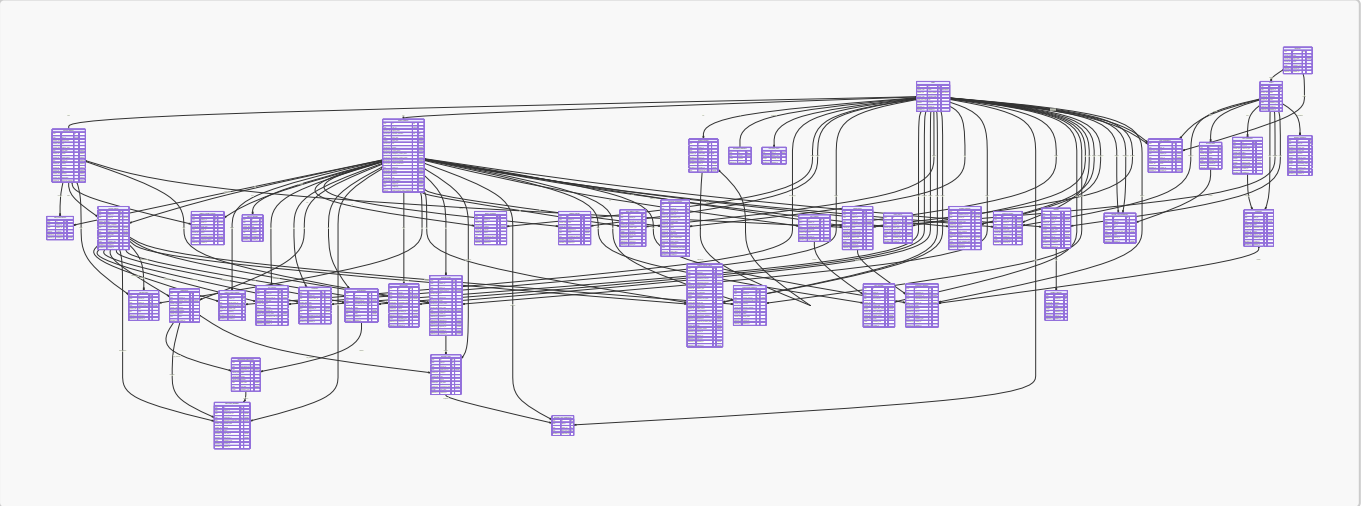
Status: working data-integrity map
Last updated: 2026-07-10
Source: `services/api/app/models.py` and Alembic migrations through `20260710_0015`

This file is a field-level ERD for implementation reviews. It is intentionally more detailed than a product overview because Dr. AI depends on one consistent source of truth for account identity, patient profile data, doctor profile data, Clinical Event context, and future commercial/admin foundations.

Legend

- `PK`: primary key.
- `FK`: foreign key.
- `UK`: unique key or unique column.
- `IDX`: indexed column.
- `nullable`: optional column.
- `jsonb`: PostgreSQL `JSONB`.
- `datetime_tz`: timezone-aware SQLAlchemy `DateTime`.
- `uuid`: PostgreSQL UUID.
- `varchar_N`: string column with length `N`.

Full Field-Level ERD



Entity Groups

Identity And Access

- `users`: account identity, credentials, active status, and primary role.
- `doctor_profiles`: doctor-owned profile fields used by registration, account editing, hospital display, and patient workspace context.
- `patient_profiles`: patient-owned identity/profile source of truth, including avatar URL, location, contact preferences, height, and weight.
- `refresh_sessions`: rotating refresh-token families and logout/session revocation state.

- `care_relationships`: the access boundary between doctors and patients.
- `role_assignments`: future hospital/clinic-scoped roles for doctor, specialist, and admin administration.

Patient And Clinical Workflow

- `clinical_events`: the care-episode object that should organize diagnoses, medications, prescriptions, reminders, imaging, recovery plans, notes, consultations, and patient updates.
- `medical_records`, `patient_allergies`, `chronic_diseases`, `visit_history`, and `diagnoses`: the patient medical-history layer.
- `medical_history_reviews`: doctor verification/correction metadata for patient-owned history without silently overwriting patient identity.
- `medications`, `prescriptions`, `prescription_medications`, and `medication_reminders`: medication-management and reminder state.
- `medical_imaging_assets`: real imaging metadata and review workflow.
- `demo_medical_assets`: historical/demo upload-presign foundation; should not be treated as the production medical-imaging source of truth.
- `recovery_plans`, `recovery_tasks`, and `recovery_task_completions`: doctor plan drafting/review plus patient Today completion tracking.
- `chat_threads`, `chat_messages`, and `chat_rate_limits`: educational chat state, safety metadata, and per-user throttling.
- `consultation_requests`: specialist consultation request/status foundation.
- `audit_events`: metadata-only audit record for sensitive workflow actions.

Commercial And Administration Foundations

- `hospitals`, `clinics`, `doctor_teams`, and `doctor_team_memberships`: future clinic administration and team structure.
- `clinic_resources`, `equipment`, `appointments`, and `equipment_bookings`: future clinic operations and scheduling/resource-management foundation.
- `invoices`, `payment_statuses`, and `revenue_summaries`: bounded billing and revenue foundation.
- `analytics_events`: operational/product analytics events.
- `support_requests`: support, deletion, and feedback queue foundation.

Integrity Rules

- `users` stores account identity and role; role-specific facts live in `doctor_profiles` or `patient_profiles`.
- Patient profile data entered during registration or onboarding must persist in `patient_profiles` when the value is first-class profile state. This includes display name, birth date/year, sex/gender, phone, emergency contact, preferred language, contact preference, avatar URL, country, province, city, home address, height, and weight.
- Medical-history facts should live in the normalized patient-history tables where possible: allergies in `patient_allergies`, chronic conditions in `chronic_diseases`, visits in `visit_history`, diagnoses in `diagnoses`, and broader documents/summaries in `medical_records`.
- Doctor profile location uses the same country/province/city reference data in registration and account editing.

- Doctor-patient access flows through `care_relationships`; doctor screens should not fetch patient records outside that relationship.
- `clinical_events` are the organizing object for care work. Medications, prescriptions, reminders, imaging, recovery plans, notes, consultations, and patient updates should link to a clinical event whenever the workflow context exists.
- Medical-history corrections and doctor verification belong in `medical_history_reviews`, not by silently overwriting patient-owned identity fields.
- Logout/session validity belongs in `refresh_sessions`; account pages should call the existing logout endpoint and clear local session state after user confirmation.
- Financial, operations, specialist, and admin tables are schema foundations. They should remain hidden from production-facing navigation until the matching workflows, authorization, validation, empty states, and tests are implemented.

Current Caveats

- Some future/commercial tables already exist in the data foundation but do not yet have complete product surfaces.
- `metadata_json` and `audit_metadata_json` are intentionally flexible, but they should not become a dumping ground for fields that need filtering, validation, editing, or cross-page synchronization.
- `synthetic_identifier` is currently the patient-share code. Product copy may call it a patient code, but the database column remains named for historical compatibility.